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Endpoint Evaluation for Sustainable Virtual Meetings

Abstract

Devices, systems, methods, and processes for conducting sustainability-aware virtual meetings are described herein. When establishing virtual meetings, each of the participants can have various devices, locations, histories, and other data associated with them. This can create various sustainable attributes at each stage of the virtual meeting. However, the endpoints of the virtual meetings, such as the client and/or server-based devices, are typically most responsible for producing negative environmental outcomes. Therefore, embodiments described herein can focus on the endpoints of virtual meetings and evaluate various endpoint attributes that are related to the sustainability attributes with each endpoint. If the endpoints are found to exceed a sustainability threshold, such as a budget or other amount, a video bit rate associated with the endpoint is reduced. This reduction can be from a high-definition signal to a standard-definition signal. The reduction may also be overridden based on either a prompt or heuristic event.

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Background/Summary

[0001] The present disclosure relates to online virtual meetings. More particularly, the present disclosure relates to changing one or more settings of a virtual meeting based on an evaluation of the endpoints of a virtual meeting.

BACKGROUND

[0002] Virtual meetings have become increasingly popular in recent years, especially since the COVID-19 pandemic made remote work and social distancing the norm. Virtual meetings can refer to any type of meeting that takes place over the internet, using video conferencing software or other tools. These tools can be operated on personal computing devices and even mobile computing devices such as smartphones, tablets, etc.

[0003] Virtual meetings have many advantages. For example, people can collaborate from anywhere in the world, which is especially helpful for teams that are distributed across different locations. Virtual meetings also save time and money by eliminating the need for travel, as well as reducing the need for physical office space. In addition, virtual meetings can be recorded, which can be useful for reviewing discussions and decisions later on.

[0004] During a virtual meeting, participants generate and transmit various types of data. This includes audio data, such as speech and background noise captured by microphones, and video data, such as images of participants and their surroundings captured by cameras. Text data, including chat messages and typed messages, is also generated, and transmitted between participants. Additionally, screen sharing data, which includes content shared from one participant's screen to others, and metadata, such as the date and time of the meeting and participant information, are also transmitted.

[0005] The video data that is transmitted during a virtual meeting includes images of the participants and their surroundings, as captured by the cameras on their devices. This data can include live video of participants speaking or listening, as well as any movements or gestures they make during the meeting. The video data may constitute a large portion of the data transmitted during a virtual meeting. As a result, many resources may be utilized to facilitate the transmission of this data. These resources can include the use of electricity generated from a variety of non-sustainable and/or relatively higher-carbon emitting power sources.

SUMMARY OF THE DISCLOSURE

[0006] Systems and methods for changing one or more settings of a virtual meeting based on an evaluation of the endpoints of a virtual meeting in accordance with embodiments of the disclosure are described herein. In some embodiments, a device, includes a processor, a memory communicatively coupled to the processor, a communication port coupled with a second device, and a virtual meeting logic. The logic is configured to establish a virtual meeting with a plurality of video streams at a first bit rate, select one or more endpoints within the virtual meeting, gather sustainability attributes associated with the one or more endpoints, determine endpoint attributes based on the sustainability attributes, compare the endpoint attributes to a predetermined sustainability threshold, establish, in response to the predetermined sustainability threshold being exceeded, a second bit rate, and reduce at least one of the plurality of video streams.

[0007] In some embodiments, the first bit rate is associated with a high-definition video stream.

[0008] In some embodiments, the second bit rate is associated with a standard-definition video stream.

[0009] In some embodiments, the sustainability attributes include at least a location of the one or more endpoints.

[0010] In some embodiments, determining the endpoint attributes include at least determining one or more negative environmental impacts associated with each endpoint.

[0011] In some embodiments, the one or more negative environmental impacts include at least greenhouse gas emissions.

[0012] In some embodiments, the virtual meeting logic is further configured to collect

sustainability data.

[0013] In some embodiments, the second bit rate is lower than the first bit rate.

[0014] In some embodiments, a device includes a processor, a memory communicatively coupled to the processor, and a virtual meeting logic. The logic is configured to optimize a virtual meeting, wherein the optimization includes establishing the virtual meeting with a plurality of video streams at a first bit rate, gathering sustainability attributes, determining endpoint attributes based on the sustainability attributes, comparing the endpoint attributes to a predetermined sustainability threshold, establishing, in response to the predetermined sustainability threshold being exceeded, a second bit rate, and reducing at least one of the plurality of video streams, monitor for a re-evaluation event, and re-optimize, in response to detecting the re-evaluation event, the virtual meeting.

[0015] In some embodiments, each of the plurality of video streams is associated with a participant within the virtual meeting.

[0016] In some embodiments, the re-evaluation event is a change in the participants of the virtual meeting.

[0017] In some embodiments, the re-evaluation event is an elapsing of a period of time.

[0018] In some embodiments, monitoring for the re-evaluation event includes at least monitoring telemetry data associated with the virtual meeting.

[0019] In some embodiments, the re-evaluation event includes a change in monitored telemetry data.

[0020] In some embodiments, the virtual meeting allows for a video bit rate to include a screen share of a computer.

[0021] In some embodiments, the re-evaluation event includes initialization of the screen share within the virtual meeting.

[0022] In some embodiments, the re-evaluation event includes ending the screen share within the virtual meeting.

[0023] In some embodiments, a method of conducting a virtual meeting includes establishing the virtual meeting with a plurality of video streams at a first bit rate, gathering sustainability attributes, determining endpoint attributes based on the sustainability attributes, comparing the endpoint attributes to a predetermined sustainability threshold, establishing, in response to the predetermined sustainability threshold being exceeded, a second bit rate, and reducing at least one of the plurality of video streams.

[0024] In some embodiments, the method further includes providing an option to engage a sustainability override during the virtual meeting.

[0025] In some embodiments, in response to engaging the sustainability override, at least one of the plurality of video streams reduced to the second bit rate are reset to the first bit rate.

[0026] Other objects, advantages, novel features, and further scope of applicability of the present disclosure will be set forth in part in the detailed description to follow, and in part will become apparent to those skilled in the art upon examination of the following or may be learned by practice of the disclosure. Although the description above contains many specificities, these should not be construed as limiting the scope of the disclosure but as merely providing illustrations of some of the presently preferred embodiments of the disclosure. As such, various other embodiments are possible within its scope. Accordingly, the scope of the disclosure should be determined not by the embodiments illustrated, but by the appended claims and their equivalents.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0027] The description of the present disclosure will be more fully understood with reference to the

following figures, which are presented as exemplary embodiments of the disclosure and should not be construed as a complete recitation of the scope of the disclosure, wherein:

[0028] FIG. 1 is a conceptual illustration of a network with network devices suitable for endpoint evaluation in accordance with various embodiments of the disclosure;

[0029] FIG. 2 is a simplified diagram of a network of devices suitable for endpoint evaluation in accordance with various embodiments of the disclosure;

[0030] FIG. 3 is a conceptual schematic diagram of a one-to-many virtual meeting in accordance with various embodiments of the disclosure;

[0031] FIG. 4 is a conceptual schematic diagram of a many-to-many virtual meeting in accordance with various embodiments of the disclosure;

[0032] FIG. 5 is a graph depicting the emissions generated at various points in a virtual meeting in accordance with various embodiments of the disclosure;

[0033] FIG. 6 is a flowchart depicting a process for conducting a virtual meeting in accordance with various embodiments of the disclosure;

[0034] FIG. 7 is a flowchart depicting a process for utilizing re-evaluation events within virtual meetings in accordance with various embodiments of the disclosure;

[0035] FIG. 8 is a flowchart depicting a process for utilizing override events within virtual meetings in accordance with various embodiments of the disclosure; and

[0036] FIG. 9 is a conceptual block diagram of a device suitable for use in a sustainability-aware virtual meeting system in accordance with various embodiments of the disclosure.

[0037] Corresponding reference characters indicate corresponding components throughout the several figures of the drawings. Elements in the several figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures might be emphasized relative to other elements for facilitating understanding of the various presently disclosed embodiments. In addition, common, but well-understood, elements that are useful or necessary in a commercially feasible embodiment are often not depicted in order to facilitate a less obstructed view of these various embodiments of the present disclosure.

DETAILED DESCRIPTION

[0038] In response to the problems described above, devices and methods are discussed herein that allow for conducting sustainability-aware virtual meetings over the network. As will be shown below, a majority of emissions associated with the endpoints of a virtual meeting, including the virtual meeting client and/or a private network center. In many embodiments, various decisions about a virtual meeting can be adjusted in response to the type of available power, with an emphasis on these endpoints.

[0039] For example, in some embodiments, when only high carbon electricity (i.e., electricity produced through means that produce a higher level of carbon pollution), a video endpoint can reduce the video bit rate, such as going from high-definition to standard-definition. In various embodiments, such a change can reduce overall energy usage by approximately twenty percent during a video call. Conversely, when low carbon or other sustainable electricity is available, the video bit rate can be increased, such as going from standard-definition to high-definition.

[0040] In certain embodiments, the endpoint can determine carbon intensity by calling an application programming interface, or otherwise utilizing a lookup method. This lookup method can be provided from third-party providers in some embodiments. When needed, the participant affected by the reduced video bit rate can be prompted that this is occurring. However, in more embodiments, the prompt may provide an option to override the reduced video bit rate, which may come in handy for scenarios where higher definition is key to the discussion occurring in the virtual meeting.

[0041] In various embodiments, participants in virtual meetings can connect from different types of endpoints (e.g., fixed vs. battery powered; browsers to huge screens; different energy source types,

etc.). However, traditional methods treat each of these locations and/or types similarly. Therefore, in various embodiments described herein, systems and methods can create an endpoint representation comprised of several types of data. This data can comprise identifying attributes related to the endpoint type, such as the physical components, power ratings, virtual or binary-like energy consuming features, codec choices, refresh speed, etc.

[0042] In this way, various services or quality levels can be adjusted based on user profiles (SD vs. HD, encrypted or not, audio vs. video, etc.). The available resources and capabilities on each device can be understood and utilized as needed to reduce energy consumption (thereby increasing the overall sustainability of the virtual meeting). The meeting profile can be associated, created, or provided by the participant hosting the meeting. The meeting profile can include multiple categories or heuristics to apply to different types of user profiles.

[0043] In additional embodiments, the virtual meetings may also examine and configure the source of electricity that is powering the devices within the network that is conducting the virtual meeting. By utilizing these methods, a virtual meeting can be conducted that can require less electricity to conduct, either by transferring less data and/or by selecting video transmission settings based on the power sources available to the network.

[0044] Aspects of the present disclosure may be embodied as an apparatus, system, method, or computer program product. Accordingly, aspects of the present disclosure may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, or the like) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “function,” “module,” “apparatus,” or “system.” Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more non-transitory computer-readable storage media storing computer-readable and/or executable program code. Many of the functional units described in this specification have been labeled as functions, in order to emphasize their implementation independence more particularly. For example, a function may be implemented as a hardware circuit comprising custom VLSI circuits or gate arrays, off-the-shelf semiconductors such as logic chips, transistors, or other discrete components. A function may also be implemented in programmable hardware devices such as via field programmable gate arrays, programmable array logic, programmable logic devices, or the like.

[0045] Functions may also be implemented at least partially in software for execution by various types of processors, logics, or controllers. An identified function of executable code may, for instance, comprise one or more physical or logical blocks of computer instructions that may, for instance, be organized as an object, procedure, or function. Nevertheless, the executables of an identified function need not be physically located together but may comprise disparate instructions stored in different locations which, when joined logically together, comprise the function and achieve the stated purpose for the function.

[0046] Indeed, a function of executable code may include a single instruction, or many instructions, and may even be distributed over several different code segments, among different programs, across several storage devices, or the like. Where a function or portions of a function are implemented in software, the software portions may be stored on one or more computer-readable and/or executable storage media. Any combination of one or more computer-readable storage media may be utilized. A computer-readable storage medium may include, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing, but would not include propagating signals. In the context of this document, a computer readable and/or executable storage medium may be any tangible and/or non-transitory medium that may contain or store a program for use by or in connection with an instruction execution system, apparatus, processor, controller, logic, or device.

[0047] Computer program code for carrying out operations for aspects of the present disclosure

may be written in any combination of one or more programming languages, including an object-oriented programming language such as Python, Java, Smalltalk, C++, C#, Objective C, or the like, conventional procedural programming languages, such as the “C” programming language, scripting programming languages, and/or other similar programming languages. The program code may execute partly or entirely on one or more of a user's computer and/or on a remote computer or server over a data network or the like.

[0048] A component, as used herein, comprises a tangible, physical, non-transitory device. For example, a component may be implemented as a hardware logic circuit comprising custom VLSI circuits, gate arrays, or other integrated circuits; off-the-shelf semiconductors such as logic chips, transistors, or other discrete devices; and/or other mechanical or electrical devices. A component may also be implemented in programmable hardware devices such as field programmable gate arrays, programmable array logic, programmable logic devices, or the like. A component or element may comprise one or more silicon integrated circuit devices (e.g., chips, die, die planes, packages) or other discrete electrical devices, in electrical communication with one or more other components through electrical lines of a printed circuit board (PCB) or the like. Each of the functions and/or modules described herein, in certain embodiments, may alternatively be embodied by or implemented as a component.

[0049] A circuit, as used herein, comprises a set of one or more electrical and/or electronic components providing one or more pathways for electrical current. In certain embodiments, a circuit may include a return pathway for electrical current, so that the circuit is a closed loop. In another embodiment, however, a set of components that does not include a return pathway for electrical current may be referred to as a circuit (e.g., an open loop). For example, an integrated circuit may be referred to as a circuit regardless of whether the integrated circuit is coupled to ground (as a return pathway for electrical current) or not. In various embodiments, a circuit may include a portion of an integrated circuit, an integrated circuit, a set of integrated circuits, a set of non-integrated electrical and/or electrical components with or without integrated circuit devices, or the like. In one embodiment, a circuit may include custom VLSI circuits, gate arrays, logic circuits, or other integrated circuits; off-the-shelf semiconductors such as logic chips, transistors, or other discrete devices; and/or other mechanical or electrical devices. A circuit may also be implemented as a synthesized circuit in a programmable hardware device such as field programmable gate array, programmable array logic, programmable logic device, or the like (e.g., as firmware, a netlist, or the like). A circuit may comprise one or more silicon integrated circuit devices (e.g., chips, die, die planes, packages) or other discrete electrical devices, in electrical communication with one or more other components through electrical lines of a printed circuit board (PCB) or the like. Each of the functions and/or modules described herein, in certain embodiments, may be embodied by or implemented as a circuit.

[0050] Further, as used herein, reference to reading, writing, storing, buffering, and/or transferring data can include the entirety of the data, a portion of the data, a set of the data, and/or a subset of the data. Likewise, reference to reading, writing, storing, buffering, and/or transferring non-host data can include the entirety of the non-host data, a portion of the non-host data, a set of the non-host data, and/or a subset of the non-host data.

[0051] Aspects of the present disclosure are described below with reference to schematic flowchart diagrams and/or schematic block diagrams of methods, apparatuses, systems, and computer program products according to embodiments of the disclosure. It will be understood that each block of the schematic flowchart diagrams and/or schematic block diagrams, and combinations of blocks in the schematic flowchart diagrams and/or schematic block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor, controller, and/or logic of a computer or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor, controller, logic, or other programmable data processing apparatus, create means for implementing the functions

and/or acts specified in the schematic flowchart diagrams and/or schematic block diagrams block or blocks.

[0052] It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. Other steps and methods may be conceived that are equivalent in function, logic, or effect to one or more blocks, or portions thereof, of the illustrated figures. Although various arrow types and line types may be employed in the flowchart and/or block diagrams, they are understood not to limit the scope of the corresponding embodiments. For instance, an arrow may indicate a waiting or monitoring period of unspecified duration between enumerated steps of the depicted embodiment.

[0053] In the following detailed description, reference is made to the accompanying drawings, which form a part thereof. The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description. The description of elements in each figure may refer to elements of proceeding figures. Like numbers may refer to like elements in the figures, including alternate embodiments of like elements.

[0054] Reference throughout this specification to “one embodiment,” “an embodiment,” or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present disclosure. Thus, appearances of the phrases “in one embodiment,” “in an embodiment,” and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment, but mean “one or more but not all embodiments” unless expressly specified otherwise. The terms “including,” “comprising,” “having,” and variations thereof mean “including but not limited to”, unless expressly specified otherwise. An enumerated listing of items does not imply that any or all of the items are mutually exclusive and/or mutually inclusive, unless expressly specified otherwise. The terms “a,” “an,” and “the” also refer to “one or more” unless expressly specified otherwise.

[0055] Lastly, the terms “or” and “and/or” as used herein are to be interpreted as inclusive or meaning any one or any combination. Therefore, “A, B or C” or “A, B and/or C” mean “any of the following: A; B; C; A and B; A and C; B and C; A, B and C.” An exception to this definition will occur only when a combination of elements, functions, steps, or acts are in some way inherently mutually exclusive.

[0056] Referring to FIG. 1, a conceptual illustration of a network **100** with network devices powered suitable for sustainability-aware virtual meetings in accordance with various embodiments of the disclosure is shown. In many embodiments, the network **100** may comprise a plurality of devices that are configured to transmit and receive data related to providing, recording, and processing virtual meetings (i.e., online collaborations). In various embodiments, virtual meeting service servers **110** are connected to a wide-area network such as, for example, the Internet **120**. In further embodiments, virtual meeting service servers **110** can be configured to transmit a variety of data across the Internet **120** to any number of computing devices such as, but not limited to, personal computers **130**, virtual presentation devices **140**, and mobile computing devices including laptop computers **170**, cellular phones **160**, portable tablet computers **180** and wearable computing devices **190**. In additional embodiments, virtual meeting data may be mirrored or otherwise supplemented in additional cloud-based service provider servers or edge network systems. In still additional embodiments, the virtual meeting service servers **110** can be hosted as virtual servers within a cloud-based service.

[0057] In further embodiments, the sending and receiving of virtual meeting data can occur over the Internet **120** through wired and/or wireless connections. In the embodiment depicted in FIG. 1, the mobile computing devices are connected wirelessly to the Internet **120** via a wireless network

access point **150**. It should be understood by those skilled in the art that the types of wired and/or wireless connections between devices on the network **100** can be comprised of any combination of devices and connections as needed.

[0058] In various embodiments, the network **100** may broadly accept virtual meeting data, such as, but not limited to audio and video data from users via personal computers **130**, virtual presentation devices **140**, and/or mobile computing devices. These embodiments may, based on one or more thresholds or other network conditions/configurations, that video data may need to be transmitted to very other device within a virtual meeting. This may require determining the quantity of video transmissions within the virtual meeting. In more embodiments, the video being transmitted within the video data can utilize a reduced size video transmission data rate to transmit video frames at a lower rate than the standard frame rate, thereby reducing the amount of data that needs to be transmitted and processed over the Internet **120**.

[0059] As discussed in more detail below, virtual meetings can be established that have a certain configuration. Based upon this configuration, various sustainability attributes data can be collected. Various features and/or configurations can be selected that increase the overall sustainability of the virtual meeting. These changes can be monitored throughout the meeting. Based on these changes during a virtual meeting, the configurations can change either automatically or in response to a participant being prompted to change one or more configurations or settings.

[0060] Although a specific embodiment for a network **100** suitable for utilizing a reduced size video transmission data rate within a virtual meeting is described above with respect to FIG. **1**, any of a variety of systems and/or processes may be utilized in accordance with embodiments of the disclosure. For example, the virtual meeting service servers **110** may be configured to receive data and transfer it to another device within the network **100**. In this way, it may be configured as a relay of the data. However, in some embodiments, the selection of data rate can be done on one or more user devices. The elements depicted in FIG. **1** may also be interchangeable with other elements of FIGS. **2-9** as required to realize a particularly desired embodiment.

[0061] Referring to FIG. **2**, a simplified diagram of a network of devices suitable for virtual meetings powered by various power source types in accordance with various embodiments of the disclosure is shown. The network **200** can include a plurality of devices, e.g., routers, which can be in communication with each other and/or a remote server, which may be connected to the network via a router **220** (R5). The network **200** depicted in FIG. **2** is shown as a simplified, conceptual network where a router **210** (R1) is connected to a router **230** (R2) which may route data to router **250** (R3) or router **240** (R4) onward to router **220** (R5). Those skilled in the art will understand that a network **200** can include a large variety of devices and be arranged in a virtually limitless number of combinations based on the desired application and available deployment environment.

[0062] Additionally, it is recognized that the terms “power” and “energy” are often used interchangeably in many colloquial settings but have distinct differences. Specifically, energy is accepted as the capacity of a system or device to do work (such as in kilowatt-hours (kWh)), while power is the rate at which energy is transferred (often in watts (W)). Power represents how fast energy is being used or produced. With this in mind, it should be understood that various elements of the present disclosure may utilize common terms like “power lines,” “power grids,” power source,” “power consumption,” and “power plant” when describing energy delivery and utilization, even though those skilled in the art will recognize that those elements are delivering or processing energy (specifically electricity) at a certain rate of power. References to these terms are utilized herein specifically to increase the ease of reading.

[0063] Traditionally, devices operating within a network **200** have not considered various aspects of operation that can relate to the overall sustainability of the network. For example, devices in communication networks have often used grid-supplied energy as a primary power source. This grid-supplied energy can regularly provide energy that has been generated by a non-sustainable or negative environmental impacts-heavy power source such as a coal-powered power plant.

However, modern power grids often have more diverse and cleaner energy sources for the provided generated energy. Some devices can still be powered by power sources that utilize fossil fuels, such as the router **240 (R4)** as depicted in FIG. 2. Alternatively, some devices can operate by using renewable sources of energy, such as the router **250 (R3)** which is conceptually depicted as being powered by solar power.

[0064] Those skilled in the art will recognize that the generation of electricity within the various power plants often creates some pollution or, more generally, one or more negative environmental impacts, which can often come in the form of emissions. However, these negative environmental impacts can come in a variety of forms including, but not limited to, land use, ozone depletion, ozone formation inhibition, acidification, eutrophication (freshwater, marine, and terrestrial), abiotic resource depletion (minerals, metals, and fossil fuels), toxicity, water use, negative soil quality change, ionizing radiation, hazardous waste creation, etc. As such, these negative environmental impact measurements can be measured with specific units to quantify these changes. Various aspects of energy use can be associated with one or more of these negative environmental impacts and classified as one or more sustainability attributes.

[0065] In the embodiment depicted in FIG. 2, the operation of a non-sustainable coal-powered power plant will create a sizeable amount of negative environmental impacts in the form of carbon emissions and the like. Contrast that with a solar array which may not create emissions when generating electricity, but may negative environmental impacts, such as carbon emission generation, associated with the production and/or disposal of the solar array. Various methods of measuring these negative environmental impacts may occur. One measurement may be to examine the waste products created by the power generated (such as nuclear waste, vs. solar array e-waste, etc.).

[0066] Another measurement of negative environmental impacts that can be utilized when comparing power sources is to determine the amount of greenhouse or carbon emissions released per unit of electricity generated. Specifically, various embodiments described herein may utilize the CO₂e kg/kWh metric which measure the amount of kilowatt hours produced per kilogram of carbon dioxide gases released into the environment. Therefore, when discussing a negative environmental impacts-heavy power source compared to a clean (er) power source, the clean power source can, for example, have a better CO₂e kg/kWh rating compared to the negative environmental impacts-heavy power source. Utilizing a cleaner power source thus provides for a more sustainable network operation.

[0067] In order to maximize the overall sustainability of a network, it may be desirable to increase the use of cleaner power sources with a lower overall negative environmental impact as opposed to power sources with a higher overall negative environmental impact when operating the network. Thus, there can be a need to be aware of the source of energy provided at each device along the route of data travel. Additionally, other factors such as the attributes unique to each device can be factored in, along with the current and/or expected traffic, etc. Once known, an optimal method of traversing the data may need to be calculated. As discussed in more detail, this path algorithm can be utilized to better optimize the locations selected within a network for data travel.

[0068] Other methods may be utilized to increase sustainability in network operations. In many embodiments, the network devices themselves may have one or more features or other capabilities that can allow for a more efficient operation. For example, a network router may be operated in a lower power mode or be powered off entirely for a specific period of time or until an event occurs. Additional embodiments may utilize various other power-saving capabilities that can be turned on or off remotely or in response to a subsequent event or predetermined threshold being exceeded. Often, operations performed by the network devices can be utilized in scenarios where network performance will not be affected or is affected such that no loss in user experience occurs. By utilizing less power during operation, a higher level of sustainability can be achieved.

[0069] Together, the type of power source providing electricity to a network device, along with the

various sustainability-related capabilities of the router can be understood as the sustainability attributes of that network device. During operation, one or more devices within the network may seek and collect the sustainability attributes of various network devices, which can provide insight into both the type of power source providing power to the device, but also the various capabilities of the network device that may be activated to provide more efficient operation.

[0070] Additionally, when generating various scores, metrics, or other evaluations of the network devices within a network **200**, the sustainability attributes can vary based on a variety of factors such as the time of day, current network traffic, expected network traffic, and historical usage patterns. For example, a network router may receive energy from a solar power source during the day but receives energy from a coal-powered power plant at night. In these instances, an averaged score may be used, or a unique score may be generated at the time of operation. In another example, network traffic may be such that removing one or more network devices from the optimal sustainable data paths may negatively affect user experiences, such as when a sporting event occurs. As such, scores may be generated at numerous times depending on the desired application. Often, the act of measurement may negatively affect sustainability such that determining the proper amount of measurements for a given outcome may be determined.

[0071] Although a specific embodiment for a network **200** is described above with respect to FIG. 2, any of a variety of systems and/or processes may be utilized in accordance with embodiments of the disclosure. For example, the network could be broken into a plurality of partitions, wherein each partition could have specific needs, service level agreements, etc. that can alter sustainability-optimization. The elements depicted in FIG. 2 may also be interchangeable with other elements of FIGS. 1 and 3-9 as required to realize a particularly desired embodiment.

[0072] Referring to FIG. 3, a conceptual schematic diagram of a one-to-many virtual meeting **300** in accordance with various embodiments of the disclosure is shown. In some virtual meetings, it can be configured such that one (or a few) people are the main subjects of the meeting and that many other attendees will be primarily watching them. During these one-to-many meetings, there is a majority of dialogue from the presenter to the attendees.

[0073] In the embodiment depicted in FIG. 3, the presenter may utilize their personal computing device, such as a first laptop **320** to capture and generate a video transmission (conceptually shown as the subject captured in frame). The virtual meeting can be facilitated by a virtual meeting service **310** that can connect one or more attendees together. The presenter may connect to the virtual meeting service over a network through a plurality of network devices (conceptually shown as router R6). During the presentation of the virtual meeting, the video transmission **330** can be transmitted to the various attendees, such as to a second laptop **340** (through a plurality of network devices conceptually shown as router R7), a smartphone **360** (through a plurality of network devices conceptually shown as router R8), a tablet **350** (through a plurality of network devices conceptually shown as router R9), a desktop computer (through a plurality of network devices conceptually shown as router R10), and a third laptop **380** (through a plurality of network devices conceptually shown as router R11).

[0074] During the virtual meeting, the video transmission **330** can be shown on each of the network-connected devices of the attendees. However, to reduce the amount of data and/or power that is required, one or more configurations or settings on the devices can be changed. Thus, the virtual meeting may be conducted in a more sustainable way.

[0075] Referring to FIG. 4, a conceptual schematic diagram of a many-to-many virtual meeting **400** in accordance with various embodiments of the disclosure is shown. In some virtual meetings, it can be configured such that everyone can be the main subject of the meeting depending on who is speaking. During these many-to-many meetings, no attendee necessarily has a majority of dialogue during the meeting.

[0076] In the embodiment depicted in FIG. 4, the attendees **420-490** may utilize their available computing devices, such as, but not limited to, laptop computers, desktop computers, smartphones,

tablets, and/or presentation devices to capture and generate a video transmission. This may require determining the quantity of video transmissions within the virtual meeting. These computing devices can each be connected to a virtual meeting service **310** through a plurality of network devices conceptually shown as routers **R12-R19**. The virtual meeting service **310** can connect one or more attendees together.

[0077] Similar to the virtual meeting depicted in FIG. 3, the video transmission can be shown on each of the network-connected devices of the attendees **420-490**. However, to reduce the amount of data and/or power that is required to be transmitted by the virtual meeting service **310**, one or more capabilities, features, or video bit rates can be enabled and/or disabled. In some embodiments, a user may be prompted to change devices. For example, user **490** may be prompted to join an audio only virtual meeting via a phone which typically requires far less power to join the same meeting.

[0078] Although specific embodiments for virtual meetings are described above with respect to FIGS. 3-4, any of a variety of systems and/or processes may be utilized in accordance with embodiments of the disclosure. For example, the virtual meetings may be conducted in a peer-to-peer configuration wherein the processes carried out by the virtual meeting service **310** are instead conducted on one or more of the client applications within the computing devices of the attendees. The elements depicted in FIGS. 3-4 may also be interchangeable with other elements of FIGS. 1-2 and 5-9 as required to realize a particularly desired embodiment.

[0079] Referring to FIG. 5, a graph **500** depicting the emissions generated at various points in a virtual meeting in accordance with various embodiments of the disclosure is shown. Virtual meetings can utilize various devices for conducting the meeting. In many embodiments, an example virtual meeting may include a pair of endpoints, such as a virtual meeting client (i.e., a virtual meeting app on a user computing device), or a network device that is part of a private data center, often associated with the virtual meeting provider. In some embodiments, the virtual meeting can have a number of different endpoints, wherein multiple people are on the call and therefore incorporate a greater number of virtual meeting clients.

[0080] Other devices can often be implicated or otherwise utilized during a virtual meeting or call. Often, a participant in the virtual meeting is using a computing device, such as a smart phone, tablet, or laptop, that can connect through a wireless home network. Similarly, other home network equipment such as routers, etc. may be used. In additional embodiments, one or more operations associated with the virtual call may be executed or implicate aspects of a public cloud service. In more embodiments, the virtual meeting backbone network devices can also be utilized during a virtual meeting.

[0081] However, as can be seen in the embodiment depicted in FIG. 5, the overwhelming share of greenhouse gas (GHG) emissions during a sample thirty-minute high-definition one-to-one video call is due to either the virtual meeting clients **510**, or the private data center **550**. Specifically, the virtual meeting clients **510** utilize sixty-six percent while the private data center **550** utilizes thirty-two percent. The remaining roughly two percent is caused by the public cloud **520** and virtual meeting backbone network **540**. The emissions caused by a home internet network **530** is negligible. In various embodiments, the virtual meeting configuration may exclude a private data center, thus confirming the largest sources of carbon-based emissions are from the virtual meeting clients **510**. It is with this insight, that embodiments described herein can focus on the endpoints of these virtual meetings for analysis (the virtual meeting clients **510** and the private data center **550**).

[0082] Although a specific embodiment for a graph **500** depicting the emissions generated at various points in a virtual meeting is described above with respect to FIG. 5, any of a variety of systems and/or processes may be utilized in accordance with embodiments of the disclosure. For example, the specific mix of emissions may vary based on the composition of the call, the length, the number of participants, etc. The values here are shown for illustration purposes and are not meant to be limiting. As those skilled in the art will recognize, the specific amount of emissions could be different based on different factors, etc. (e.g., a user has a different sized monitor, or uses

a higher power source at home). Additionally, other negative environmental effects may occur beyond greenhouse gas emissions, which can also be accounted for when evaluating sustainability attributes of a virtual meeting. The elements depicted in FIG. 5 may also be interchangeable with other elements of FIGS. 1-4 and 6-9 as required to realize a particularly desired embodiment.

[0083] Referring to FIG. 6, a flowchart depicting a process 600 for conducting a virtual meeting in accordance with various embodiments of the disclosure is shown. In many embodiments, the process 600 can gather virtual meeting settings (block 610). These settings can be associated with various aspects of a virtual meeting including, but not limited to, the number of participants, the number of endpoints, the type of endpoints, the attributes and/or capabilities of the endpoints, the power sources of the endpoints, various sustainability metrics, data bit rates, etc. These settings can be gathered by each endpoint device or may be gathered by a centralized device or service.

[0084] In a number of embodiments, the process 600 can determine sustainability attributes (block 620). As previously discussed, the sustainability attributes can be determined based on third-party services or by other stored data related to various endpoint devices, users, and/or locations. The sustainability attributes can include, but are not limited to, power source type, device type, power usage, etc.

[0085] In more embodiments, the process 600 can evaluate end point attributes (block 630). As discussed above, the end point attributes may include data associated with the type of end point device, the amount of power used, historical data related to previous virtual meetings, projected power usage during an upcoming virtual meeting, and/or the sustainability capabilities of the end point. Those skilled in the art will recognize that other attributes may be determined based on the application desired.

[0086] In further embodiments, the process 600 can determine if the meeting exceeds a sustainability threshold (block 635). The sustainability threshold may be a projected budget associated with the virtual meeting. In some embodiments, the threshold may be a static threshold set on one or more factors. In certain embodiments, the threshold may be dynamic in nature and be based on changing factors such as overall energy usage and/or the mix of energy source types.

[0087] If it is determined that a sustainability threshold has not been exceeded, certain embodiments of the process 600 can set a video data rate to a high-definition bit rate (block 640). The high-definition bit rate can be configured to be any bit rate that can sustain a high-definition video feed within the virtual meeting. However, the bit rate may be configured to be above a minimum baseline bit rate level associated with a given quality of service required or other restriction.

[0088] However, if it is determined that a sustainability threshold has been exceeded, then some embodiments of the process 600 can set a video data rate to a standard-definition bit rate (block 650). The standard-definition bit rate can be configured to be any bit rate that can sustain a standard-definition (or conversely a non-high-definition bit rate) video feed within the virtual meeting. However, the bit rate may be configured to be below a minimum baseline bit rate level in order to maintain certain issues or other disruptions within a network connection.

[0089] In still more embodiments, the process 600 can conduct the virtual meeting (block 660). In some embodiments, the virtual meeting can be conducted in a static matter insofar as once the video bit rate settings are determined, they remain static during the duration of the virtual meeting. However, in more embodiments, the process 600 may dynamically reassess or adjust the video data rate in response to new information. As those skilled in the art will recognize, the virtual meeting can include a plurality of video streams at various video data bit rates. These streams can include videos of the participants, avatars, icons, and/or screen shares.

[0090] Although a specific process for conducting virtual meetings is described above with respect to FIG. 6, any of a variety of systems and/or processes may be utilized in accordance with embodiments of the disclosure. For example, the process 600 may be carried out within a centralized virtual meeting service or may be carried out on a virtual meeting client software, such

as the software associated with the virtual meeting host. The elements depicted in FIG. 6 may also be interchangeable with other elements of FIGS. 1-5 and 7-9 as required to realize a particularly desired embodiment.

[0091] Referring to FIG. 7, a flowchart depicting a process **700** for utilizing re-evaluation events within virtual meetings in accordance with various embodiments of the disclosure is shown. In many embodiments, the process **700** can determine a sustainability budget (block **710**). A sustainability budget can be determined based on a number of input factors. This determination can be generated by one or more machine learning process. However, in some embodiments, the determination may be based on one or more heuristic processes. The sustainability budget determination factors can include various aspects related to the virtual meeting.

[0092] In a number of embodiments, the process **700** can evaluate end point attributes (block **720**). Again, as discussed above, the end point attributes may include data associated with the type of end point device, the amount of power used, historical data related to previous virtual meetings, projected power usage during an upcoming virtual meeting, and/or the sustainability capabilities of the end point. Those skilled in the art will recognize that other attributes may be determined based on the application desired.

[0093] In more embodiments, the process **700** determine if the sustainability budget has been exceeded (block **735**). The sustainability budget may be a threshold value associated with the virtual meeting. In some embodiments, the budget may be a static threshold set on one or more factors. In certain embodiments, the budget may be dynamic in nature and be based on changing factors such as overall energy usage and/or the mix of energy source types.

[0094] If it is determined that a sustainability threshold has not been exceeded, certain embodiments of the process **700** can set a video data rate to a high-definition bit rate (block **740**). The high-definition bit rate can be configured to be any bit rate that can sustain a high-definition video feed within the virtual meeting. However, the bit rate may be configured to be above a minimum baseline bit rate level associated with a given quality of service required or other restriction.

[0095] However, if it is determined that a sustainability threshold has been exceeded, then some embodiments of the process **700** can set a video data rate to a standard definition bit rate (block **750**). The standard-definition bit rate can be configured to be any bit rate that can sustain a standard-definition (or conversely a non-high-definition bit rate) video feed within the virtual meeting. However, the bit rate may be configured to be below a minimum baseline bit rate level in order to maintain certain issues or other disruptions within a network connection.

[0096] In additional embodiments, the process **700** can further determine if a re-evaluation event occurred (block **755**). A re-evaluation event can be configured as any event or change associated with the virtual meeting. For example, in some embodiments, the adding or subtracting of participants and their associated video streams can affect the overall sustainability of the virtual meeting, triggering a re-evaluation. In more embodiments, the initialization or ending of a screen share can be a re-evaluation event. In yet more embodiments, the re-evaluation event can comprise a change in monitored telemetry data associated with the virtual meeting. Those skilled in the art will recognize that a variety of different changes can be configured as a trigger for a re-evaluation, which is often configured to re-optimize the virtual meeting. If it is determined that a re-evaluation event has occurred, the process **700** can again evaluate end point attributes (block **720**).

[0097] However, if it is determined that a re-evaluation event has not occurred, then the process **700** can, in various embodiments, conduct the virtual meeting (block **760**). In some embodiments, the virtual meeting can be conducted for a period of time before a further evaluation occurs. For example, during the virtual meeting, in still further embodiments, the process **700** can determine if the virtual meeting has ended (block **765**). If it is determined that the virtual meeting has not ended, certain embodiments of the process **700** can continue to determine if a re-evaluation event has occurred (block **755**). However, if it is determined that the virtual meeting has ended, the process

700 itself may end.

[0098] Although a specific process **700** for utilizing re-evaluation events within virtual meetings is described above with respect to FIG. 7, any of a variety of systems and/or processes may be utilized in accordance with embodiments of the disclosure. For example, the process **700** can conduct the re-evaluation after a period of time (having the elapsed time being the re-evaluation event). The elements depicted in FIG. 7 may also be interchangeable with other elements of FIGS. **1-6** and **8-9** as required to realize a particularly desired embodiment.

[0099] Referring to FIG. 8, a flowchart depicting a process **800** for utilizing override events within virtual meetings in accordance with various embodiments of the disclosure is shown. In certain optional embodiments, the process **800** can determine a sustainability budget (block **810**). A sustainability budget can be determined based on a number of input factors. This determination can be generated by one or more machine learning process. However, in some embodiments, the determination may be based on one or more heuristic processes. The sustainability budget determination factors can include various aspects related to the virtual meeting.

[0100] In a number of embodiments, the process **800** can evaluate end point attributes (block **820**). Once again, as discussed above, the end point attributes may include data associated with the type of end point device, the amount of power used, historical data related to previous virtual meetings, projected power usage during an upcoming virtual meeting, and/or the sustainability capabilities of the end point. Those skilled in the art will recognize that other attributes may be determined based on the application desired.

[0101] In more embodiments, the process **800** can determine an exceeded sustainability budget (block **830**). The sustainability budget may be a threshold value associated with the virtual meeting. In some embodiments, the budget may be a static threshold set on one or more factors. In certain embodiments, the budget may be dynamic in nature and be based on changing factors such as overall energy usage and/or the mix of energy source types.

[0102] In additional embodiments, the process **800** can set a video data rate to a lower bit rate (block **840**). The standard-definition bit rate can be configured to be any bit rate that can sustain a standard-definition (or conversely a non-high-definition bit rate) video feed within the virtual meeting. However, the bit rate may be configured to be below a minimum baseline bit rate level in order to maintain certain issues or other disruptions within a network connection.

[0103] In further embodiments, the process **800** can conduct a virtual meeting (block **850**). Virtual meetings can be conducted for a period of time and/or until an event occurs. In still more embodiments, the process **800** can determine if a sustainability override option has been engaged or otherwise executed (block **855**). The override option can be configured as an interface or other selectable option to a user or virtual meeting host that can indicate to the process **800** that a higher video bit rate is desired.

[0104] In various embodiments, if it has been determined that an override option has not been engaged, the process **800** can again continue to conduct the virtual meeting (block **850**). However, if an override option has been executed, some embodiments of the process **800** can set a video data rate to a higher bit rate (block **860**). The higher video bit rate may be configured to allow for high-definition video, but may be configured to marginally increase the video bit rate, which can allow for the user or virtual meeting host to select the override option again and/or until the video bit rate is sufficient for their purposes.

[0105] In still further embodiments, the process **800** can continue to conduct the virtual meeting. In some embodiments, the virtual meeting can be conducted in a static matter from this point forward insofar as once the video bit rate settings are determined, they remain static during the duration of the virtual meeting. However, in more embodiments, the process **800** may dynamically reassess or adjust the video data rate in response to new information or override deselection(s).

[0106] For example, the process **800** can determine if the override option has ended (block **875**). In certain embodiments, an override, once selected, can be utilized to increase the video bit rate for

the duration of the virtual meeting. However, in some embodiments, the override may only be activated for a period of time, or may be deactivated by the user. If the user override option has not ended, the process **800** may continue to conduct the virtual meeting (block **870**). Conversely, if the override option has ended, various embodiments of the process **800** can return or otherwise reset to set the video data rate to a lower bit rate, up to and including standard-definition video bit rates or an equivalent. (block **840**).

[0107] Although a specific process for conducting a virtual meeting on a client device is described above with respect to FIG. **8**, any of a variety of systems and/or processes may be utilized in accordance with embodiments of the disclosure. For example, the process **800** can have the override selected virtually by one or more machine learning processes upon determining that certain contents or characteristics of the virtual meeting deem it necessary and/or desired, such as to maintain a predetermined quality of service level. The elements depicted in FIG. **8** may also be interchangeable with other elements of FIGS. **1-7** and **9** as required to realize a particularly desired embodiment.

[0108] Referring to FIG. **9**, a conceptual block diagram of a device suitable for use in a sustainability-aware virtual meeting system in accordance with various embodiments of the disclosure is shown. The embodiment of the conceptual block diagram depicted in FIG. **9** can illustrate a conventional server computer, workstation, desktop computer, laptop, tablet, network appliance, e-reader, smartphone, or other computing device, and can be utilized to execute any of the applications and/or logic components presented herein. The device **900** may, in some examples, correspond to physical devices or to virtual resources described herein.

[0109] In many embodiments, the device **900** may include an environment **902** such as a baseboard or “motherboard,” in physical embodiments that can be configured as a printed circuit board with a multitude of components or devices connected by way of a system bus or other electrical communication paths. Conceptually, in virtualized embodiments, the environment **902** may be a virtual environment that encompasses and executes the remaining components and resources of the device **900**. In more embodiments, one or more processor(s) **904**, such as, but not limited to, central processing units (CPUs), controllers, etc. that can be configured to operate in conjunction with a chipset **906**. The processor(s) **904** can be standard programmable CPUs that perform arithmetic and logical operations necessary for the operation of the device **900**.

[0110] In additional embodiments, the processor(s) **904** can perform one or more operations by transitioning from one discrete, physical state to the next through the manipulation of switching elements that differentiate between and change these states. Switching elements generally include electronic circuits that maintain one of two binary states, such as flip-flops, and electronic circuits that provide an output state based on the logical combination of the states of one or more other switching elements, such as logic gates. These basic switching elements can be combined to create more complex logic circuits, including registers, adders-subtractors, arithmetic logic units, floating-point units, and the like.

[0111] In certain embodiments, the chipset **906** may provide an interface between the processor(s) **904** and the remainder of the components and devices within the environment **902**. The chipset **906** can provide an interface to a random-access memory (RAM **908**), which can be used as the main memory in the device **900** in some embodiments. In a number of embodiments, the memory can be communicatively coupled to the processor(s) **904** to carry out one or more instructions. The chipset **906** can further be configured to provide an interface to a computer-readable storage medium such as a read-only memory (ROM **910**) or non-volatile RAM (NVRAM) for storing basic routines that can help with various tasks such as, but not limited to, starting up the device **900** and/or transferring information between the various components and devices. The ROM **910** or NVRAM can also store other application components necessary for the operation of the device **900** in accordance with various embodiments described herein.

[0112] Different embodiments of the device **900** can be configured to operate in a networked

environment using logical connections to remote computing devices and computer systems through a network, such as the network **940**. The chipset **906** can include functionality for providing network connectivity through a network interface controller **912**, which may provide access to external devices through, for example, a plurality of communication ports such as, but not limited to, a network interface card (NIC). A NIC may comprise a gigabit Ethernet adapter or similar component. The plurality of communication ports can be capable of connecting or otherwise coupling the device **900** to other devices over the network **940**. It is contemplated that multiple communication ports, such as NICs may be present in the device **900**, connecting the device to other types of networks and remote systems.

[0113] In further embodiments, the device **900** can be connected to a storage **918** that provides non-volatile storage for data accessible by the device **900**. The storage **918** can, for example, store an operating system **920**, applications **922**, and data, which are described in greater detail below. The storage **918** can be connected to the environment **902** through a storage controller **914** connected to the chipset **906**. In certain embodiments, the storage **918** can consist of one or more physical storage units. The storage controller **914** can interface with the physical storage units through a serial attached SCSI (SAS) interface, a serial advanced technology attachment (SATA) interface, a fiber channel (FC) interface, or other type of interface for physically connecting and transferring data between computers and physical storage units.

[0114] The device **900** can store data within the storage **918** by transforming the physical state of the physical storage units to reflect the information being stored. The specific transformation of physical state can depend on various factors. Examples of such factors can include, but are not limited to, the technology used to implement the physical storage units, whether the storage **918** is characterized as primary or secondary storage, and the like.

[0115] For example, the device **900** can store information within the storage **918** by issuing instructions through the storage controller **914** to alter the magnetic characteristics of a particular location within a magnetic disk drive unit, the reflective or refractive characteristics of a particular location in an optical storage unit, or the electrical characteristics of a particular capacitor, transistor, or other discrete component in a solid-state storage unit, or the like. Other transformations of physical media are possible without departing from the scope and spirit of the present description, with the foregoing examples provided only to facilitate this description. The device **900** can further read or access information from the storage **918** by detecting the physical states or characteristics of one or more particular locations within the physical storage units.

[0116] In addition to the storage **918** described above, the device **900** can have access to other computer-readable storage media to store and retrieve information, such as program modules, data structures, or other data. It should be appreciated by those skilled in the art that computer-readable storage media is any available media that provides for the non-transitory storage of data and that can be accessed by the device **900**. In some examples, the operations performed by a cloud computing network, and or any components included therein, may be supported by one or more devices similar to device **900**. Stated otherwise, some or all of the operations performed by the cloud computing network, and or any components included therein, may be performed by a device **900** or a plurality of devices operating in a cloud-based arrangement.

[0117] By way of example, and not limitation, computer-readable storage media can include volatile and non-volatile, removable, and non-removable media implemented in any method or technology. Computer-readable storage media includes, but is not limited to, RAM, ROM, erasable programmable ROM (“EPROM”), electrically-erasable programmable ROM (“EEPROM”), flash memory or other solid-state memory technology, compact disc ROM (“CD-ROM”), digital versatile disk (“DVD”), high definition DVD (“HD-DVD”), BLU-RAY, or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to store the desired information in a non-transitory fashion.

[0118] As mentioned briefly above, the storage **918** can store an operating system **920** utilized to

control the operation of the device **900**. According to one embodiment, the operating system comprises the LINUX operating system. According to another embodiment, the operating system comprises the WINDOWS® SERVER operating system from MICROSOFT Corporation of Redmond, Washington. According to further embodiments, the operating system can comprise the UNIX operating system or one of its variants. It should be appreciated that other operating systems can also be utilized. The storage **918** can store other system or application programs and data utilized by the device **900**.

[0119] In various embodiment, the storage **918** or other computer-readable storage media is encoded with computer-executable instructions which, when loaded into the device **900**, may transform it from a general-purpose computing system into a special-purpose computer capable of implementing the embodiments described herein. These computer-executable instructions may be stored as application **922** and transform the device **900** by specifying how the processor(s) **904** can transition between states, as described above. In some embodiments, the device **900** has access to computer-readable storage media storing computer-executable instructions which, when executed by the device **900**, perform the various processes described above with regard to FIGS. **1-8**. In more embodiments, the device **900** can also include computer-readable storage media having instructions stored thereupon for performing any of the other computer-implemented operations described herein.

[0120] In still further embodiments, the device **900** can also include one or more input/output controllers **916** for receiving and processing input from a number of input devices, such as a keyboard, a mouse, a touchpad, a touch screen, an electronic stylus, or other type of input device. Similarly, an input/output controller **916** can be configured to provide output to a display, such as a computer monitor, a flat panel display, a digital projector, a printer, or other type of output device. Those skilled in the art will recognize that the device **900** might not include all of the components shown in FIG. **9** and can include other components that are not explicitly shown in FIG. **9**, or might utilize an architecture completely different than that shown in FIG. **9**.

[0121] As described above, the device **900** may support a virtualization layer, such as one or more virtual resources executing on the device **900**. In some examples, the virtualization layer may be supported by a hypervisor that provides one or more virtual machines running on the device **900** to perform functions described herein. The virtualization layer may generally support a virtual resource that performs at least a portion of the techniques described herein.

[0122] In many embodiments, the device **900** can include a virtual meeting logic **924**, that can be configured to perform one or more of the various steps, processes, operations, and/or other methods that are described above. Often, the virtual meeting logic **924** can be a set of instructions stored within a non-volatile memory that, when executed by the processor(s) **904** can carry out these steps, etc. In some embodiments, the virtual meeting logic **924** may be a client application that resides on a network-connected device, such as, but not limited to, a personal or mobile computing device. In these embodiments, the virtual meeting logic **924** can facilitate the establishment and conduct one or more virtual meetings within the device, such as those devices depicted in FIGS. **1** and **3-4**. In certain embodiments, the virtual meeting logic **924** may be configured to receive and process configuration suggestions that can direct one or more configurations, such as energy-saving settings to be either enabled or disabled.

[0123] In a number of embodiments, the storage **918** can include configuration data **928**. As described above, configuration data **928** can be utilized to optimize the sustainability of a virtual meeting. In some embodiments, the configuration data **928** can indicate the type of virtual meeting that will be conducted, as well as the types of devices that are likely to attend. The configuration data **928** may also include, in certain embodiments, the network paths, devices, or types of connections that may be utilized during a virtual meeting. In still more embodiments, the configuration data **928** can include various energy-saving or other features that can be enabled and/or disabled.

[0124] In still more embodiments, the storage **918** may include meeting profile data **930**. Also described above, meeting profile data **930** can be generated upon receiving the user profile data **929** from each of the participants. However, in some embodiments, that may not be necessary and the meeting profile data **930** can be generated from a partial selection of user profile data **929**. The meeting profile data **930** may include data related to the overall configurations available for the devices associated with the user profile data **929** received. Indeed, in various embodiments, the meeting profile data **930** can allow for the determination of an optimal configuration of settings throughout the virtual meeting in order to maximize the overall sustainability of the virtual meeting.

[0125] In still more embodiments, the storage **918** can include sustainability data **932**. Sustainability data can include sustainability attributes of various devices, elements, and other components of a virtual meeting network. The sustainability data **932** may comprise both capability data related to various devices but can also include the power source type associated with each device within a network proposed or being utilized by a virtual meeting. In some embodiments, the sustainability data **932** can include historical data such that decisions or inferences can be generated without all current real-time data, or to make a prediction of upcoming network conditions.

[0126] Finally, in many embodiments, data may be processed into a format usable by one or more machine-learning (ML) model(s) **926** (e.g., feature vectors), and or other pre-processing techniques. The ML model(s) **926** may be any type of ML model, such as any of various supervised models, reinforcement models, and/or unsupervised models. The ML model(s) **926** may include one or more of linear regression models, logistic regression models, decision trees, Naïve Bayes models, neural networks, k-means cluster models, random forest models, and/or other types of ML model(s) **926**.

[0127] The ML model(s) **926** can be configured to generate inferences to make predictions or draw conclusions from data. An inference can be considered the output of a process of applying a model to new data. This can occur by learning from historical data and use that learning to predict future outcomes. These predictions are based on patterns and relationships discovered within the data. To generate an inference, the trained model can take input data and produce a prediction or a decision. The input data can be in various forms, such as images, audio, text, or numerical data, depending on the type of problem the model was trained to solve. The output of the model can also vary depending on the problem, and can be a single number, a probability distribution, a set of labels, a decision about an action to take, etc.

[0128] The ML model(s) **926** may be configured to predict the usage and/or configurations of a better sustainability-optimized virtual meeting. Likewise, the ML model(s) **926** may be configured in certain embodiments to process various data to make a determination on whether a configuration may need to be changed. In some embodiments, the ML model(s) **926** may be utilized to determine which specific configuration needs to be adjusted.

[0129] Although a specific embodiment for a device suitable for carrying out the various steps, processes, methods, and operations described above is discussed with respect to FIG. **9**, any of a variety of systems and/or processes may be utilized in accordance with embodiments of the disclosure. For example, the device may be virtualized or disposed as a logic and data stores within a mobile or personal general computing device, allowing for the transformation of the device, upon execution by a processor/controller, to a sustainability-aware virtual meeting device. The elements depicted in FIG. **9** may also be interchangeable with other elements of FIGS. **1-8** as required to realize a particularly desired embodiment.

[0130] Although the present disclosure has been described in certain specific aspects, many additional modifications and variations would be apparent to those skilled in the art. In particular, any of the various processes described above can be performed in alternative sequences and/or in parallel (on the same or on different computing devices) in order to achieve similar results in a manner that is more appropriate to the requirements of a specific application. It is therefore to be

understood that the present disclosure can be practiced other than specifically described without departing from the scope and spirit of the present disclosure. Thus, embodiments of the present disclosure should be considered in all respects as illustrative and not restrictive. It will be evident to the person skilled in the art to freely combine several or all of the embodiments discussed here as deemed suitable for a specific application of the disclosure. Throughout this disclosure, terms like “advantageous”, “exemplary” or “example” indicate elements or dimensions which are particularly suitable (but not essential) to the disclosure or an embodiment thereof and may be modified wherever deemed suitable by the skilled person, except where expressly required. Accordingly, the scope of the disclosure should be determined not by the embodiments illustrated, but by the appended claims and their equivalents.

[0131] Any reference to an element being made in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” All structural and functional equivalents to the elements of the above-described preferred embodiment and additional embodiments as regarded by those of ordinary skill in the art are hereby expressly incorporated by reference and are intended to be encompassed by the present claims.

[0132] Moreover, no requirement exists for a system or method to address each and every problem sought to be resolved by the present disclosure, for solutions to such problems to be encompassed by the present claims. Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. Various changes and modifications in form, material, workpiece, and fabrication material detail can be made, without departing from the spirit and scope of the present disclosure, as set forth in the appended claims, as might be apparent to those of ordinary skill in the art, are also encompassed by the present disclosure.

Claims

1. A device, comprising: a processor; a memory communicatively coupled to the processor; a communication port coupled with a second device; and a virtual meeting logic configured to: establish a virtual meeting with a plurality of video streams at a first bit rate; select one or more endpoints within the virtual meeting; gather sustainability attributes associated with the one or more endpoints; determine endpoint attributes based on the sustainability attributes; compare the endpoint attributes to a predetermined sustainability threshold; establish, in response to the predetermined sustainability threshold being exceeded, a second bit rate; and reduce at least one of the plurality of video streams.
2. The device of claim 1, wherein the first bit rate is associated with a high-definition video stream.
3. The device of claim 1, wherein the second bit rate is associated with a standard-definition video stream.
4. The device of claim 1, wherein the sustainability attributes include at least a location of the one or more endpoints.
5. The device of claim 1, wherein determining the endpoint attributes include at least determining one or more negative environmental impacts associated with each endpoint.
6. The device of claim 5, wherein the one or more negative environmental impacts comprise at least greenhouse gas emissions.
7. The device of claim 6, wherein the virtual meeting logic is further configured to collect sustainability data.
8. The device of claim 1, wherein the second bit rate is lower than the first bit rate.
9. A device, comprising: a processor; a memory communicatively coupled to the processor; and a virtual meeting logic configured to: optimize a virtual meeting, wherein the optimization comprises: establishing the virtual meeting with a plurality of video streams at a first bit rate; gathering sustainability attributes; determining endpoint attributes based on the sustainability

attributes; comparing the endpoint attributes to a predetermined sustainability threshold; establishing, in response to the predetermined sustainability threshold being exceeded, a second bit rate; and reducing at least one of the plurality of video streams; monitor for a re-evaluation event; and re-optimize, in response to detecting the re-evaluation event, the virtual meeting.

10. The device of claim 9, wherein each of the plurality of video streams is associated with a participant within the virtual meeting.

11. The device of claim 10, wherein the re-evaluation event is a change in the participants of the virtual meeting.

12. The device of claim 9, wherein the re-evaluation event is an elapsing of a period of time.

13. The device of claim 9, wherein monitoring for the re-evaluation event comprises at least monitoring telemetry data associated with the virtual meeting.

14. The device of claim 13, wherein the re-evaluation event comprises a change in monitored telemetry data.

15. The device of claim 13, wherein the virtual meeting allows for a video bit rate to comprise a screen share of a computer.

16. The device of claim 15, wherein the re-evaluation event comprises initialization of the screen share within the virtual meeting.

17. The device of claim 15, wherein the re-evaluation event comprises ending the screen share within the virtual meeting.

18. A method of conducting a virtual meeting, comprising: establishing the virtual meeting with a plurality of video streams at a first bit rate; gathering sustainability attributes; determining endpoint attributes based on the sustainability attributes; comparing the endpoint attributes to a predetermined sustainability threshold; establishing, in response to the predetermined sustainability threshold being exceeded, a second bit rate; and reducing at least one of the plurality of video streams.

19. The method of claim 18, wherein the method further comprises providing an option to engage a sustainability override during the virtual meeting.

20. The method of claim 19, wherein, in response to engaging the sustainability override, at least one of the plurality of video streams reduced to the second bit rate are reset to the first bit rate.
